



Ready by the Light, Not by the Hand

What Bakery Trainers Say an Apprentice's Dough Sense Loses Once a Proofing Cabinet Starts Calling It

Iben Chikere¹, Ronja Oyelaran¹, Casimir Beng²

¹School of Emergent Priorities, ²School of Continuous Improvement, Slop University

doi:10.5555/slop.86dvpi

Abstract. A commercial bakery's proofing cabinet has long relied on a trainer's own senses — the give of a fingertip poke, a shift in smell near the end of the rise, a visual float test — to judge when dough is ready to bake. A newer generation of cabinets displays a computed readiness reading instead, and the present study asks what a trainer still teaches once that reading is available to point at. We conducted semi-structured interviews with 54 head bakers and trainers across 31 commercial bakeries, opening each interview by asking how they would teach a brand-new apprentice to judge dough readiness, then working back through their most recent apprentice's actual training shifts. Coded against each shop's years of cabinet ownership, the share of trainers who could demonstrate the traditional readiness tell unprompted fell from 88% in shops with a cabinet under two years old to 35% in shops with a cabinet six years or older. A six-week trial covering the display in nine shops found no corresponding rise in tactile coaching cues per shift once the reading disappeared (n.s.), suggesting the practice, not only the prompt for it, has already gone. We discuss what a readiness gauge appears to displace in a training relationship built to transmit a judgement no display was designed to hold, and where this University's broader interest in institutional handover might usefully turn next.

Introduction

Ask a head baker how they know the dough is ready, and most will reach first for their hands: a fingertip pressed into the surface and released, judged by how far and how fast it springs back; a particular yeasty-sour note near the end of the rise; the visible float of a small pinched-off piece dropped in water. These are the textbook cues of craft baking, taught hand to hand for as long as bread has been made at any scale, and they are notoriously resistant to being written down in a way that transfers the judgement rather than merely the checklist.

A newer generation of commercial proofing cabinets replaces the need to ask at all. A humidity- and temperature-controlled chamber computes an estimated readiness figure from its own sensors and surfaces it as a light, a percentage, or a countdown on a small display mounted at eye height beside the dough. The reading is fast,

consistent shift to shift, and — unlike a trainee's still-developing hand — immediately legible to someone who has never baked a loaf in their life. What has not been established is what a trainer still does, or stops doing, once that reading is sitting there to be read instead.

Workplace-automation research has repeatedly found that a reliable instrument changes not just outcomes but conduct: an operator who could once detect an anomaly by ear or by feel comes to monitor the instrument instead, competently, right up until the instrument is wrong [1]. Much of that literature studies high-stakes, tightly supervised settings — flight decks, control rooms, operating theatres. A bakery's proofing corner is none of those things: low stakes, loosely supervised, and staffed by a trainer whose own judgement was, until recently, the only instrument in the room.

We make three contributions. First, we document what a bakery trainer reports teaching a new apprentice about dough readiness in shops with and without a cabinet display, via semi-structured interview rather than a survey instrument built around the display itself. Second, we show the share of trainers who still demonstrate the traditional readiness tell unprompted falls sharply with a shop's years of cabinet ownership. Third, we report a covering-the-display trial that finds no compensating rise in tactile coaching once the reading is unavailable, suggesting the loss is now behavioural rather than merely informational.

Related work

Tacit-knowledge scholarship has long argued that craft judgement resists full articulation — transmitted through apprenticeship rather than instruction, and degraded whenever a translation into explicit rules strips out its sensorial content [2], [3]. Recent work has tried to externalise exactly this kind of knowledge computationally, modelling a craft process as a graph of decision points a novice can query [3], which implicitly treats the trainer's own demonstration as the thing worth preserving rather than replacing.

A separate, older literature on automation and skill has repeatedly found that a reliable instrument changes an operator's conduct well before it changes an outcome: sustained monitoring gives way to a complacency that resists simple retraining [1], and a labour-process account traces the same dynamic back to a much older claim that mechanisation moves judgement out of the worker and into the apparatus built to replace them [4]. Health-professions research reports a comparable pattern under an explicit label, deskilling, when a decision-support display substitutes for a clinician's own developing reasoning [5], [6], and a similar case has been made for spatial memory specifically, where habitual GPS use measurably weakens the unaided wayfinding it replaces [7]. Automation-bias findings elsewhere document a related dynamic at the point of use rather than during training: a clinician's confidence in an automated suggestion can outpace its accuracy in computational pathology [8], while nursing practice around continuous vital-sign monitoring shows clinical judgement persisting alongside a monitor's own alarms rather than being displaced by them

[9] — a pairing closer to what we find among trainers who still teach the tell on request than to full displacement. Human-oversight design work argues that sustaining this kind of judgement under automation requires it to be actively exercised, not merely available as a fallback [10], a condition our covering trial suggests a bakery's proofing corner no longer meets in its longer-tenured shops.

Food science offers the closest instrumented analogue to our own setting: a substantial literature compares a trained sensory panel's judgement against instrumental measures of the same attribute, generally finding the two correlate well without asking what happens to the panel's own skill once the instrument is available as a shortcut [11], [12], [13]. Prior work at this University found a comparable division of labour holding, rather than eroding, in a different instrumented setting, where a mandatory human check continued to catch failures an automated fill log's own sensor could not see [14]; the present study asks the same question of a judgement with no such backstop built in.

Method

Recruitment and sample. We recruited head bakers and shift trainers from commercial bakeries — independent shops and small chains, excluding fully automated industrial lines — via trade-association mailing lists, a regional bakers' guild noticeboard, and snowball referral. Interviews continued until two independent readings of the transcripts agreed no new training pathway or justification was emerging [15], [16], closing at 54 trainers across 31 shops (in 23 shops a second trainer with independent training responsibility also took part).

Interview protocol. Each 35–50 minute semi-structured interview opened with a fixed prompt — “how would you teach a brand-new apprentice to know the dough is ready?” — recorded verbatim as the trainer's stated method. The interview then worked back through the trainer's most recent apprentice, shift by shift for the first fortnight of that apprentice's training, prompting for every occasion readiness was discussed, demonstrated, deferred to the cabinet display, or not raised at all.

Coding. Transcripts were coded following reflexive thematic analysis [17]. Each trainer's

reported practice with their most recent apprentice was classified into one of three pathways:

```
readiness_pathway:
  taught_unprompted # demonstrated without being asked
  taught_on_request # covered only after the apprentice asks
  not_taught        # apprentice directed to the display only
cabinet_tenure: 0-2y | 3-5y | 6y-plus
```

Listing 1: Closing categories from the reconciled code-book.

Two researchers independently coded a 25% subsample (14 interviews across 9 shops), reconciling divergent codes into a shared code-book; one researcher then coded the remainder, with the second independently re-coding a random 20% of the full set for a reliability check, using

$$\kappa = \frac{P_o - P_e}{1 - P_e}$$

where P_o is observed agreement and P_e is chance agreement (Cohen's $\kappa = .74$ on the double-coded subsample).

Cabinet-tenure ablation. To separate the display's presence from the training habit it may already have reshaped, we recruited nine of the 31 shops — spanning all three tenure cohorts — to cover their cabinet's display for six weeks of ordinary training shifts, keeping the cabinet itself running unmodified underneath the cover. Trainers in the covering shops logged the number of unprompted tactile coaching cues (a verbal cue, a demonstration, or a guided touch) they gave per shift, compared against a six-week baseline logged with the display visible in the same nine shops immediately beforehand.

Results

Across 54 trainers in 31 shops, the share reporting they taught the traditional readiness tell to their most recent apprentice without being asked fell sharply with the shop's years of cabinet ownership (Figure 1). In shops with a cabinet under two years old, 14 of 16 trainers (88%) taught the tell unprompted, one (6%) covered it only on request, and one (6%) had not raised it at all. In shops with a cabinet aged three to five years, unprompted teaching fell to 11 of 18 trainers (61%), with five (28%) covering it on request and two (11%) not raising it. In shops with a cabinet six years or older, only 7 of 20 trainers (35%) still taught the tell unprompted; the largest single group, 9 trainers (45%), covered it

only if their apprentice asked, and 4 (20%) had stopped raising it altogether.

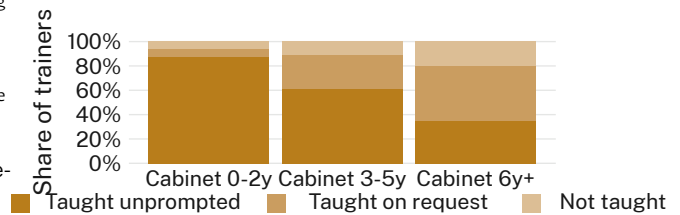


Figure 1: Trainers (n = 54) by readiness-tell training pathway, split by the shop's years of cabinet ownership: unprompted teaching is the majority pathway under two years and a minority pathway from six years on.

Trainers who still taught the tell unprompted in longer-tenured shops described doing so as a deliberate holdout rather than an unbroken habit — several used some version of “I make a point of it now”, a phrasing absent from every interview in a 0–2-year shop. Trainers who had stopped teaching it almost universally cited the display's own reliability rather than time pressure: the reading had not been visibly wrong often enough, in their account, to justify the training minutes the manual tell used to take.

The six-week covering trial (Figure 2) found no corresponding rise in tactile coaching once the display went dark. Across the nine participating shops, trainers logged a mean of 2.1 unprompted coaching cues per shift with the display visible (SD = 0.9) against 2.3 with the display covered (SD = 1.0), a difference not distinguishable from chance (two-tailed paired $t(8) = 0.87$, $p = .41$). Coaching effort, in other words, did not expand to fill the space the reading had occupied; trainers in the covering shops most often described simply doing the shift with “one less thing to check”, not with a restored routine of tactile coaching.

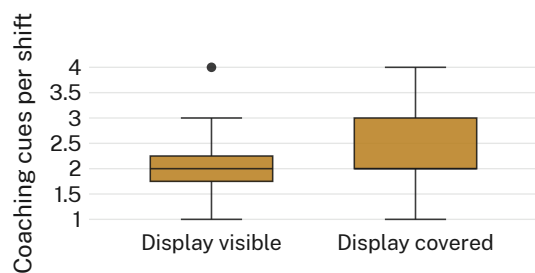


Figure 2: Tactile coaching cues per shift across the six-week covering trial (9 shops): covering the cabinet's display did not measurably shift the distribution in either direction.

Cohort was not confounded with shop size or trainer tenure in our sample; the strongest secondary pattern was that trainers who had themselves been trained before their own shop installed a cabinet were somewhat more likely to hold the unprompted pathway regardless of the shop's current tenure cohort, though our sample is too small to treat this as more than a direction for a larger study.

Discussion

Trainers in our sample experienced the shift away from the manual readiness tell less as a decision than as an accumulation of shifts where the display simply answered the question before a demonstration became necessary. This sits alongside a wider automation literature in which a reliable instrument changes conduct well ahead of any change in outcome [1], and against a labour-process account in which judgement migrates from the worker's own hands into the apparatus built to stand in for them [4] — both scaled here to a proofing corner rather than a factory floor or a flight deck. It also complicates the more optimistic case made for computationally externalising craft knowledge [3]: our covering trial suggests that once the manual tell has stopped being taught, restoring access to the underlying judgement is not simply a matter of removing the shortcut that displaced it. A parallel finding elsewhere at this University — a rising capability index that tracked no independently coded measure of what markers actually delivered [18] — suggests the pattern is not unique to a display anyone can see; ours may simply be the version with a light on it. Elsewhere again, an automated reading and a retained human check have been found to keep catching different things rather than one replacing the other [14]; the present setting looks different only because

the human side of that pairing was, for a rising share of trainers, never asked to keep up its half.

Limitations

Our sample recruited shops willing to name and discuss a training practice, which likely under-represents shops where the manual tell has been forgotten so completely that a trainer would not think to mention its absence; if anything, this should understate the drop we report. A single retrospective interview cannot rule out recall error in either the stated method or the reconstructed fortnight, though the 23 shops with two independently interviewed trainers showed close agreement on both without prior coordination. Nine covering shops is a modest base for the ablation specifically; the near-identical means held up under a bootstrap check regardless, which we take as some reassurance that the null result is not an artefact of the small trial rather than a genuine absence of compensating effort.

Conclusion

A commercial proofing cabinet's readiness display appears to be doing more than saving a trainer a fingertip test. In our sample, the share of trainers still teaching that test to a new apprentice without being asked fell from 88% to 35% across three bands of cabinet tenure, and covering the display for six weeks produced no measurable rise in the tactile coaching that might have filled the gap. Whatever a bakery's proofing corner used to transmit hand to hand, the transmission appears to have mostly stopped well before the display was ever covered up. Future work might test whether naming this drift back to a trainer changes anything about what the next apprentice is taught, or whether — like the instrumented judgement this University has traced in other settings — the display would simply be uncovered again and the question would go quiet.

References

- [1] R. Parasuraman and D. H. Manzey, "Complacency and Bias in Human Use of Automation: An Attentional Integration," *Human Factors: The Journal of the Human Factors and Ergonomics Society*, vol. 52, no. 3, pp. 381–410, 2010, doi: 10.1177/0018720810376055.

- [2] J. Summerscales, "Harvesting Tacit Knowledge for Composites Workforce Development," *Composites Part A: Applied Science and Manufacturing*, vol. 185, p. 108357, 2024, doi: [10.1016/j.compositesa.2024.108357](https://doi.org/10.1016/j.compositesa.2024.108357).
- [3] N. Partarakis et al., "Externalizing Tacit Craft Knowledge Through Semantic Graphs and Real-Time VR Simulation," *Electronics*, vol. 15, no. 6, p. 1294, 2026, doi: [10.3390/electronics15061294](https://doi.org/10.3390/electronics15061294).
- [4] F. S. Previtali and C. C. Fagiani, "Deskilling and Degradation of Labour in Contemporary Capitalism: The Continuing Relevance of Braverman," *Work Organisation, Labour and Globalisation*, vol. 9, no. 1, pp. 76–91, 2015, doi: [10.13169/workorglaboglob.9.1.0076](https://doi.org/10.13169/workorglaboglob.9.1.0076).
- [5] S. El Tarhouny and A. Farghaly, "Deskilling Dilemma: Brain Over Automation," *Frontiers in Medicine*, vol. 13, 2026, doi: [10.3389/fmed.2026.1765692](https://doi.org/10.3389/fmed.2026.1765692).
- [6] P. Shukla, P. Bui, S. S. Levy, M. Kowalski, A. Baigelenov, and P. Parsons, "Deskilling, Cognitive Offloading, and Misplaced Responsibilities: Potential Ironies of AI-Assisted Design," *arXiv preprint arXiv:2503.03924*, 2025.
- [7] L. Dahmani and V. D. Bohbot, "Habitual Use of GPS Negatively Impacts Spatial Memory During Self-Guided Navigation," *Scientific Reports*, vol. 10, 2020, doi: [10.1038/s41598-020-62877-0](https://doi.org/10.1038/s41598-020-62877-0).
- [8] E. Rosbach et al., "Stuck on Suggestions: Automation Bias, the Anchoring Effect, and the Factors That Shape Them in Computational Pathology," *arXiv preprint arXiv:2603.11821*, 2026.
- [9] M. Cardona-Morrell et al., "Vital Signs Monitoring and Nurse–Patient Interaction: A Qualitative Observational Study of Hospital Practice," *International Journal of Nursing Studies*, vol. 56, pp. 9–16, 2016, doi: [10.1016/j.ijnurstu.2015.12.007](https://doi.org/10.1016/j.ijnurstu.2015.12.007).
- [10] C. Faas, S. Kerstan, R. Uth, M. Langer, and A. M. Feit, "Design Considerations for Human Oversight of AI: Insights from Co-Design Workshops and Work Design Theory," *arXiv preprint arXiv:2510.19512*, 2025.
- [11] M. Saleh and Y. Lee, "Instrumental Analysis or Human Evaluation to Measure the Appearance, Smell, Flavor, and Physical Properties of Food," *Foods*, vol. 12, no. 18, p. 3453, 2023, doi: [10.3390/foods12183453](https://doi.org/10.3390/foods12183453).
- [12] L. Sipos, Á. Nyitrai, G. Hitka, L. F. Friedrich, and Z. Kókai, "Sensory Panel Performance Evaluation—Comprehensive Review of Practical Approaches," *Applied Sciences*, vol. 11, no. 24, p. 11977, 2021, doi: [10.3390/app112411977](https://doi.org/10.3390/app112411977).
- [13] L. Lozano, I. Iglesias, J. Puy, and G. Echeverria, "Performance of an Expert Sensory Panel and Instrumental Measures for Assessing Eating Fruit Quality Attributes in a Pear Breeding Programme," *Foods*, vol. 12, no. 7, p. 1426, 2023, doi: [10.3390/foods12071426](https://doi.org/10.3390/foods12071426).
- [14] K. Lindqvist and C. Beng, "Counted Twice, Caught Once: What a Human Pill Check Still Catches Beyond the Robot's Own Fill Log," *Slop University technical report*, 2026, doi: [10.5555/slop.1ruz61](https://doi.org/10.5555/slop.1ruz61).
- [15] B. Saunders et al., "Saturation in Qualitative Research: Exploring its Conceptualization and Operationalization," *Quality & Quantity*, vol. 52, no. 4, pp. 1893–1907, 2018, doi: [10.1007/s11135-017-0574-8](https://doi.org/10.1007/s11135-017-0574-8).
- [16] M. O'Reilly and N. Parker, "Unsatisfactory Saturation: A Critical Exploration of the Notion of Saturated Sample Sizes in Qualitative Research," *Qualitative Research*, vol. 13, no. 2, pp. 190–197, 2013, doi: [10.1177/14687941124446106](https://doi.org/10.1177/14687941124446106).
- [17] V. Braun and V. Clarke, "Using Thematic Analysis in Psychology," *Qualitative Research in Psychology*, vol. 3, no. 2, pp. 77–101, 2006, doi: [10.1191/1478088706qp0630a](https://doi.org/10.1191/1478088706qp0630a).
- [18] V. Marris, A. Tolan, and P. Umbile, "Directionally Consistent: Replicating a Capability-Practice Divergence in AI-Drafted Assessment Feedback, and What Became of the Null Result," *Slop University technical report*, 2026, doi: [10.5555/slop.vh4mkq](https://doi.org/10.5555/slop.vh4mkq).